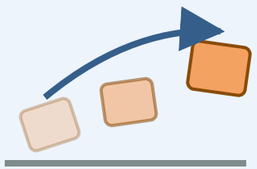
DB0: prediction diagnostic -> three Core contracts -> physics and replanning extensions

P0 Forecast



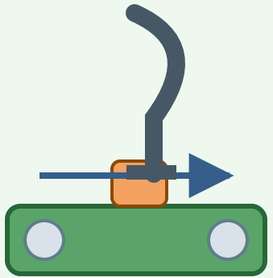
robot remains still

8 future SE(3) poses

50, ..., 400 ms

exact-current histories

T1 Belt Capture

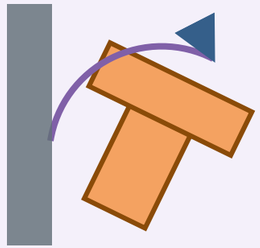


intercept -> grasp

lift 5 cm -> hold

persistent physical motion

T1+ SO(3) / OCC

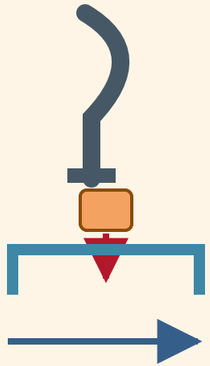


matched XYZ trajectory

independent yaw

physical hidden interval

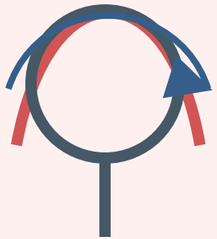
T2 Moving Goal



release into future pose

containment + dwell

T3 Rotating Buzz

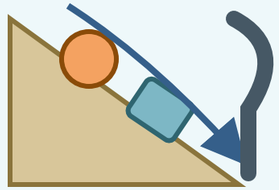


phase-aligned traversal

zero wire contact

Phase Core; Full demo

T4 Ramp Capture

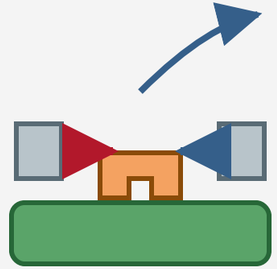


rolling / sliding / slip

gravity + friction

unsupported capture

T5 Impact Replan



one hidden branch impact

observe -> correct -> grasp

shared pre-event identity

Common physical clock

first observation = first allowed action; waiting, compute, queueing, actuation, contact, and dwell consume time

Report separately

SR | TC | CT | safety | forecast | calibration | action age | failures